

Exercise 2: Approximation and Complexity Theory

Lecturer: Aurelien Lucchi

Teaching Assistants: Nadezhda Ilieva, Jim Zhao

Problem 1 (Stone-Weierstrass theorem):

The following is a generalization of Weierstrass theorem presented in the lecture.

Theorem 1 (Stone-Weierstrass, see Theorem 2.2. in Telgarsky (2021)). *Let function class $\mathcal{F}$ of real-valued functions defined on $[0, 1]^d$ be given as follows:*

- i) *Each $f \in \mathcal{F}$ is continuous.*
- ii) *For every $\mathbf{x} \in [0, 1]^d$, there exists $f \in \mathcal{F}$ with $f(\mathbf{x}) \neq 0$.*
- iii) *For every $\mathbf{x} \neq \mathbf{x}' \in [0, 1]^d$, there exists $f \in \mathcal{F}$ with $f(\mathbf{x}) \neq f(\mathbf{x}')$ ($\mathcal{F}$ separates points).*
- iv) *$\mathcal{F}$ is closed under multiplication and vector space operations, i.e. $\mathcal{F}$ is an algebra.*

Then $\mathcal{F}$ is a universal approximator: for every continuous $g : [0, 1]^d \rightarrow \mathbb{R}$ and $\epsilon > 0$, there exists $f \in \mathcal{F}$ with

$$|f(\mathbf{x}) - g(\mathbf{x})| \leq \epsilon, \quad \forall \mathbf{x} \in [0, 1]^d.$$

a) Consider unbounded width networks with one hidden layer:

$$\mathcal{F}_{\sigma, d, m} := \mathcal{F}_{d, m} := \{x \mapsto a^\top \sigma(Wx + b) : a \in \mathbb{R}^m, W \in \mathbb{R}^{m \times d}, b \in \mathbb{R}^m\}. \quad (1)$$

$$\mathcal{F}_{\sigma, d} := \mathcal{F}_d := \bigcup_{m \geq 0} \mathcal{F}_{\sigma, d, m}. \quad (2)$$

Using Theorem 1, show that $\mathcal{F}_{\cos, d}$ is universal, where $\cos : \mathbb{R} \rightarrow \mathbb{R}$ is the cosine function.

b) Now consider the following definitions of function classes. Check whether each of the necessary conditions in Theorem 1 hold and argue whether $\mathcal{F}$ is dense in $C[0, 1]$

- i) Let $\mathcal{F} = \text{span}\{x\} = \{f : [0, 1] \rightarrow \mathbb{R} | f \text{ linear}\}$ be the function class of all linear functions.
- ii) Let $\mathcal{F} = \{f \text{ is polynomial} \mid f(0) = 0\}$ be the function class $\mathcal{F}$ of all polynomials with no constant term.
- iii) Consider now the interval $[0, 2\pi]$. (We can always define a projection function $\phi(x) : [0, 2\pi] \rightarrow [0, 1]$, to map the interval to the unit interval.)
Let $\mathcal{F} = \{a_0 + \sum_{k=1}^n (a_k \cos(kx) + b_k \sin(kx)) \mid n \in \mathbb{N}, a_k, b_k \in \mathbb{R}\}$, the function class containing all real trigonometric polynomials.

- iv) For any subset $S \subset [0, 1]$, define the characteristic function $\mathbf{1}_S : x \mapsto \begin{cases} 1 & , x \in S \\ 0 & , x \notin S \end{cases}$.

Let

$$\mathcal{F} := \text{algebra}\{\mathbf{1}_{\{a\}} \mid a \in [0, 1]\} = \{f \mid \text{supp}(f) \text{ is finite}\}.$$

where the support $\text{supp}(f)$ of a function f is defined as $\text{supp}(f) := \{x \in [0, 1] \mid f(x) \neq 0\}$.

c) (Bonus) Can you come up with other counter examples yourself?

Problem 2 (Number of affine pieces in a ReLU network):

We will prove the lemma that bounds the number of affine pieces in a ReLU network stated in the lecture notes. Denote by $\delta_A(f)$ be the (minimum) number of affine pieces of a piece-wisely linear continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$. Show that

- a) $\delta_A(\sum_k a_k f_k + b_k) \leq \sum_k \delta_A(f_k)$, for any finite sequence of piece-wisely linear continuous functions f_k , and for any real sequences a_k and b_k .

- b) Let $g_i : \mathbb{R} \rightarrow \mathbb{R}$ denote the output of some node in layer i in a ReLU network with L layers of widths $(m_1, \dots, m_L)$ as a function of the input. Using induction on i , show that the number of affine pieces $\delta_A(g_i)$ satisfies

$$\delta_A(g_i) \leq 2^i \prod_{j < i} m_j.$$

- c) Recall that $g_L : \mathbb{R} \rightarrow \mathbb{R}$ is a ReLU network with L layers of widths $(m_1, \dots, m_L)$ such that $m = \sum_{i=1}^L m_i$. The number of affine pieces in g_L satisfies

$$\delta_A(g_L) \leq \left(\frac{2m}{L} \right)^L.$$

- d) What value of L maximizes the upper bound $\left(\frac{2m}{L} \right)^L$? Although taking this value of L gives a large value of $\delta_A(g_L)$, give a reason why in practice we usually pick a value much smaller than this.

References

Matus Telgarsky. Deep learning theory lecture notes. <https://mjt.cs.illinois.edu/dlt/>, 2021. Version: 2021-10-27 v0.0-e7150f2d (alpha).